

Low-Cost Educational MUSICAL INSTRUMENT

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This musical instrument, organ, is a simple yet engaging electronic device designed to introduce kids to sound, light, and music in a fun and interactive way. It is a simple kit designed to engage kids with fun and interactive sound generation.

The organ also demonstrates fundamental electronic principles such as circuit operation, transistor amplification, and capacitor-based sound modulation. Additionally, the hands-on approach encourages creativity and problem-solving skills. With its simple design and engaging functionality, the instrument serves as a delightful, low-cost educational toy that brings joy and learning together.

The circuit is designed to generate different tones when the switches are pressed, allowing children to entertain themselves with basic musical notes. It is a do-it-yourself (DIY) device built using basic electronic components, making it an excellent learning tool for beginners in electronics and hobbyists. Fig. 1 shows the author's prototype on a breadboard.

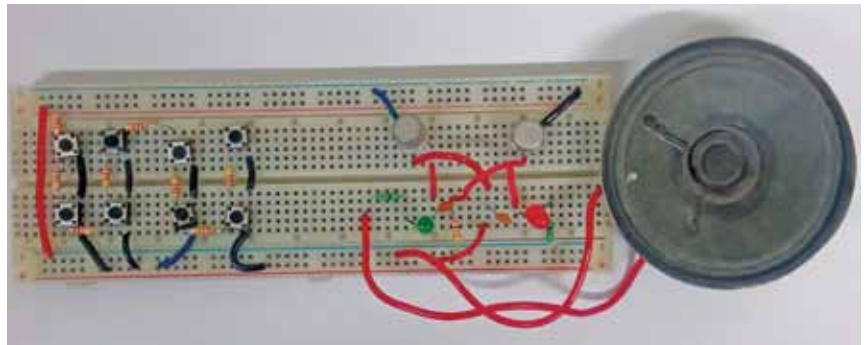


Fig. 1: Author's prototype

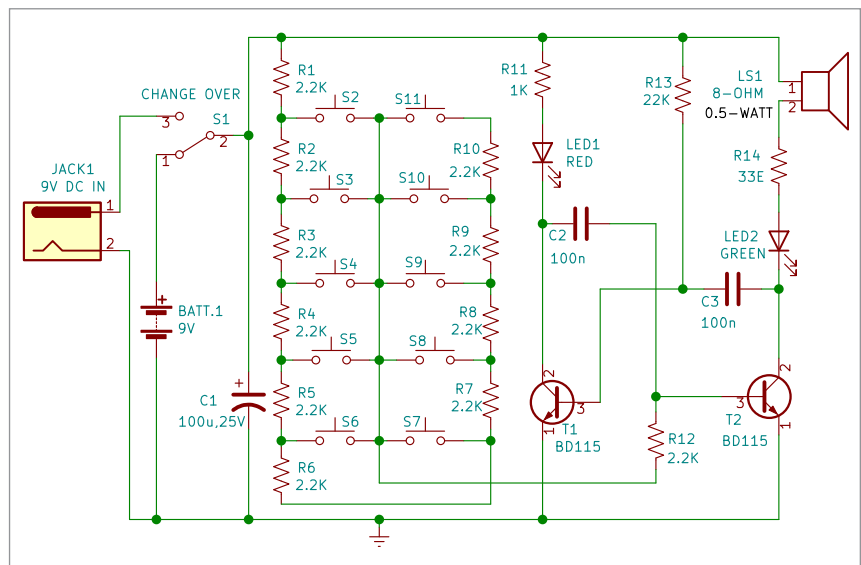


Fig. 2: Circuit diagram

Circuit and working

Fig. 2 shows the circuit diagram of the organ. It is built around a speaker, two LEDs, two BD115 transistors, a few capacitors and resistors, and ten tactile switches.

The organ produces different tones when push-to-on switches (S2 through S11) are pressed. Changeover switch S1 allows you to select between 9V DC and battery power.

You may set it to the side corresponding to your power source.

The BD115 transistors play a key role in signal amplification, ensuring that the sound produced is loud and clear. The capacitors and resistors help shape the output's tone and frequency. The LEDs function as visual indicators, lighting up when a note is played, which adds an

element of excitement for children.

The ten tactile switches act as the keys, each corresponding to a different musical tone.

When a switch (except S1) is pressed, the circuit completes the connection, allowing current to flow through the transistors and produce sound via the speaker. Each push-to-on switch (button) generates